

SADIK MONDAL

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Summary

Full-stack software engineering student who ships production-grade web applications end to end – database design, backend APIs, and interactive frontends. Built three deployed projects spanning real-time booking with database-enforced concurrency safety, mobile task-automation, and recommendation-driven media. Practices daily DSA and is preparing for GATE CSE 2027.

Professional Experience

VaultofCodes

Web Development Intern

August 2025

Remote

- Completed a focused one-month web development internship, applying full-stack development practices to real project tasks under mentorship.
- Strengthened proficiency in modern JavaScript-based web development workflows, version control, and collaborative development practices.

Forage – Tata Group

Cybersecurity Analyst Job Simulation

July 2025

Virtual

- Completed a virtual job simulation covering Identity and Access Management (IAM) fundamentals, evaluating IAM strategy against organizational security requirements.
- Designed a custom IAM solution addressing platform integration and access-control requirements, applying industry-standard security assessment methodology.

Technical Skills

Languages: C++, JavaScript (ES6+), TypeScript, Python, HTML5, CSS3

Frontend: React.js, Next.js, Tailwind CSS, Responsive Design, Component-Based Architecture

Backend: Node.js, Express.js, RESTful APIs, JWT Authentication, Socket.io

Databases: PostgreSQL, Supabase, MongoDB, MongoDB Atlas, Mongoose

Tools & Platforms: Git, GitHub, Firebase, Capacitor, Vercel, Render, Postman, VS Code

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Projects

RailVista – Railway Reservation Platform

Next.js, TypeScript, Tailwind CSS, PostgreSQL, Supabase Auth

railvista-self.vercel.app

- Engineered a full-stack booking platform modeling 100 stations, 50 trains, 577 coaches, and 37,620 individual seats with an interactive, physically accurate seat-map UI.
- Enforced booking integrity at the database layer via a PostgreSQL partial unique index on active reservations, eliminating double-booking under concurrent requests, and implemented Row-Level Security (RLS) with an automated penetration test suite for per-user data isolation.
- Cut search latency from multi-second client-side filtering to sub-second response by migrating train search to a PostgreSQL stored procedure, removing N+1 query overhead.

CleanMess – Fair Chore Rotation & Backlog Engine

React 19, TypeScript, Vite, Tailwind CSS, Capacitor, Firebase

cleanmess-app.web.app

- Designed and built a deterministic chore-rotation engine for shared living spaces, auto-assigning daily/periodic chores across four roommates on a strict, predictable rotation.
- Developed an absence-and-coverage system that reroutes chores in real time when a roommate is marked absent, while logging a capped, fairness-protected backlog to prevent unfair long-term loss of accountability.
- Compiled the web application into a native Android package using Capacitor and implemented Firebase Authentication with role-based access control (Manager vs. Roommate).

BeatMess – Personalized Music Streaming Player

React, Node.js, Vite, Tailwind CSS

beatmess-player.onrender.com

- Built a custom single-user recommendation engine that infers listening preference from implicit behavioral signals – likes, skips, partial listens, and replays – rather than manual ratings.
- Implemented an infinite auto-queue system that proactively refills playback when the upcoming buffer runs low, ensuring uninterrupted listening.
- Integrated MusicBrainz and Last.fm APIs through a rate-limited background worker for metadata enrichment and content-similarity tagging; shipped as an installable Progressive Web App.

Education

Aliah University, Kolkata

B.Tech in Computer Science & Engineering

2024 – 2028

Currently 3rd Year, 5th Semester

Ramakrishna Mission Vidyapith, Purulia

Higher Secondary (Science) – 87%

2021 – 2023

Certifications

C++ & Data Structures and Algorithms – Apna College (2025)
(2025) | Web Development Internship – VaultofCodes (2025)

Cybersecurity Analyst Job Simulation – Forage, Tata Group
Frontend Web Development – Udemy (In Progress)

Achievements

- Solved 100+ data structure and algorithm problems across LeetCode and GeeksforGeeks, spanning arrays, strings, trees, linked lists, recursion, and dynamic programming.
- Maintain a daily coding streak; expanding into graphs, tries, heaps, and segment trees while preparing for GATE CSE 2027.